

# Nathan David Obeng-Amoako, E.I.T.

Boston, MA | +1 (617) 935-7884 | [natdave545@gmail.com](mailto:natdave545@gmail.com) | <https://natdave.github.io> | [linkedin.com/in/natdave](https://www.linkedin.com/in/natdave)

## RESEARCH INTERESTS

---

- Transit Signal Priority
- Traffic Simulations / Digital Twins
- Intelligent Transportation Systems
- Multimodal Corridor Planning
- Bicycle Network Analysis
- Geospatial Methods in Transportation

## EDUCATION

---

**Doctor of Philosophy in Civil Engineering (Transportation)** Sep 2023 – Present

Northeastern University, Boston, MA (GPA: 4.00/4.00)

**Research Advisor:** Professor Peter Furth

**Master of Science in Civil Engineering (Transportation)** Sep 2023 – Apr 2025

Northeastern University, Boston, MA (GPA: 4.00/4.00)

**Bachelor of Science in Civil Engineering** Aug 2018 – Sep 2022

Kwame Nkrumah University of Science and Technology (KNUST), Kumasi

**Recognition:** Best Graduating Student, Class of 2022

## PROFESSIONAL EXPERIENCE

---

**Transportation Engineering Intern, Fehr & Peers, Oakland, CA** Apr 2026 – Jun 2026

- Supported transit signal priority and complete streets projects for state/municipal clients
- Contributed to metro station area studies and transit-oriented development analysis in the East Bay
- Developed federal grant materials for the City of Palo Alto under the OBAG 4 funding cycle
- Conducted GIS and active transportation analysis for corridor and safety planning projects

**Program Assistant, Sustainable Transportation Summer Academy, Delft, Holland** Jun 2024 – Aug 2026

- Provided logistical, academic, and evaluative support to students over three summers
- Assisted in teaching Vision Zero safety principles to 25 Northeastern University students
- Explored car-free and car-lite zone strategies to enhance active/public transportation

**Assistant Transportation Engineer, BHM Construction International UK Ltd.** Oct 2022 – Jul 2023

- Conducted over 20 quality control tests daily on construction materials to ensure strict standard compliance
- Supported site reporting and coordination with a team of 5 engineers to improve workflow efficiency

**Civil Engineering Trainee, China Henan International Corporation Ltd.** Sep 2021 – Dec 2021

- Supervised 15+ artisans daily to ensure strict adherence to project specifications and timelines
- Contributed to precise structural detailing for reinforced concrete components

## SPONSORED RESEARCH PROJECTS

---

**Project:** Digital Twin for Adaptive Traffic Signal Control **Aug 2026 – Present**

**Funding Agency:** StreetIQ, LLC (an AI-powered infrastructure intelligence platform)

- Building and calibrating a Vissim microsimulation of Boston arterial corridors as a testbed for adaptive signal control algorithm development
- Developing external signal control software via Vissim's API to enable iterative, closed-loop algorithm testing and validation against existing control

**Project:** Bicycle Level of Traffic Stress at Intersections (xLTS)

**Jun 2024 – Aug 2025**

**Funding Agency:** United States Department of Transportation (USDOT)

- Assessing the impact of high-stress crossings on bike network connectivity
- Developing algorithms and data structures for traffic stress network analysis

**Project:** Accessible Bus Stop Design in the Presence of Bike Lanes

**Sep 2023 – Jun 2024**

**Funding Agency:** Massachusetts Department of Transportation (MassDOT)

- Studied bus rider and bicyclist behavior at floating and constrained bus stops
- Proposed design improvements to reduce bus-rider-and-bicyclist conflicts

## PEER-REVIEWED JOURNAL PUBLICATIONS

---

- 2025
1. Peter Furth, Ray Saeidi-Razavi, **Nathan David Obeng-Amoako** and Milad Tahmasebi. TSP-Friendly Underlying Traffic Signal Control: An Essential Complement of Transit Signal Priority. *Future Transportation* (2025), doi: <https://doi.org/10.3390/futuretransp5040160>
  2. Joshua Kofi Asamoah, Blessing Agyei Kyem, **Nathan David Obeng-Amoako**, Armstrong Aboah, SAAM-ReflectNet: Sign-Aware Attention-Based Multitasking Framework for Integrated Traffic Sign Detection and Retroreflectivity Estimation, *Expert Systems With Applications* (2025), doi: <https://doi.org/10.1016/j.eswa.2025.128003>

### Submitted and under review:

- 2026
3. Peter G. Furth, **Nathan David Obeng-Amoako**, Armin Akhavan, Bitra Sadeghinassr, Ray Saeidi, Luis Miranda-Moreno, Christine Théberge-Barrette, Paul Moser, and Nick Ferenchak. Intersections as Barriers to Low-Stress Cycling: Intersection Level of Traffic Stress and Dual-Graph Network Modeling. *Journal of Transport Geography*.

## RESEARCH REPORTS

---

- 2024
1. Eleni Christofa, Chengbo Ai, Peter Furth, Yu-Min Yang, Dewan Tanvir Ahammed, and **Nathan David Obeng-Amoako**. (2024). *Accessible Bus Stop Design in the Presence of Bike Lanes*. Massachusetts Department of Transportation (MassDOT), Office of Transportation Planning. Report No. 24-060. Retrieved from <https://rosap.ntl.bts.gov/view/dot/79327>

## JOURNAL REVIEWS

---

- |                                                                     |                       |
|---------------------------------------------------------------------|-----------------------|
| 1. Journal of Transportation Engineering, Part A: Systems, ASCE     | <b>2026 – Present</b> |
| 2. Transportation Research Interdisciplinary Perspectives, Elsevier | <b>2025 – Present</b> |
| 3. Expert Systems with Applications, Elsevier                       | <b>2025 – Present</b> |
| 4. Transportation Research Record, SAGE Publications                | <b>2024 – Present</b> |

## APPOINTMENTS AND ADVISORY ROLES

---

- |                                                                                                                                                                                                                                                                                |                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <b>Member, Massachusetts Bicycle and Pedestrian Advisory Board, MassDOT</b>                                                                                                                                                                                                    | <b>Jan 2026 – Present</b> |
| <ul style="list-style-type: none"><li>• Appointed by Governor Maura T. Healey to advise on state-wide bicycle and pedestrian infrastructure policies and programs</li><li>• Provide guidance on transportation infrastructure, safety, and accessibility initiatives</li></ul> |                           |

## RELEVANT SOFTWARE AND SKILLS

---

- |                              |                                                                            |
|------------------------------|----------------------------------------------------------------------------|
| • <b>Python, R, MATLAB</b>   | machine learning, deep learning, advanced data analysis                    |
| • <b>PTV Vissim/Vistro</b>   | traffic flow simulations, traffic signal analysis, traffic impact analysis |
| • <b>Synchro</b>             | traffic flow modeling, traffic signal timing optimization                  |
| • <b>ArcGIS</b>              | spatial data mapping and analysis for informed decision-making             |
| • <b>Gephi</b>               | network data analysis and visualization                                    |
| • <b>AutoCAD</b>             | precision drafting and technical drawings, 2D and 3D                       |
| • <b>Bluebeam Revu</b>       | creating, editing, and marking up 2D digital blueprints                    |
| • <b>HTML/CSS/JavaScript</b> | developing web applications                                                |
| • <b>Minitab, MS Excel</b>   | data visualization, statistical computing, data organization               |

## NOTABLE DISTINCTIONS AND ACHIEVEMENTS

---

- |                                                                                        |                    |
|----------------------------------------------------------------------------------------|--------------------|
| • Grant Recipient, New England African Business Expo, Worcester, MA                    | <b>May 2025</b>    |
| • Civil & Environmental Engineering Distinguished Fellowship Award, Northeastern Univ. | <b>Aug 2023</b>    |
| • Associated Consultants Ltd. Award for Best Graduating Student, KNUST                 | <b>Jul 2023</b>    |
| • Provost Award for Best & Excellent Students in College of Engineering, KNUST (x4)    | <b>2019 – 2022</b> |
| • Academic Excellence Award, Ghana Engineering Students' Association, Ghana            | <b>Sep 2022</b>    |
| • Global Finalist, Unilever Future Leaders League Business Pitch Competition, U.K.     | <b>Jul 2022</b>    |
| • Scholar, Tullow Oil Scholarship Scheme, Ghana                                        | <b>Apr 2019</b>    |
| • Semifinalist, National Science and Maths Quiz (NSMQ), Ghana                          | <b>Jul 2018</b>    |

## LEADERSHIP EXPERIENCE

---

### **The Ckodon Foundation:**

- |                                                                                                                                                                                                                                    |                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>❑ Executive Director of Partnerships</b>                                                                                                                                                                                        | <b>May 2026 – Present</b>  |
| <ul style="list-style-type: none"><li>• Lead partnership development with universities, mentors, and sponsor organizations</li><li>• Deliver graduate admissions and test-preparation webinars to prospective applicants</li></ul> |                            |
| <b>❑ Co-Director for the Graduate Mentorship Program</b>                                                                                                                                                                           | <b>Aug 2025 – Apr 2026</b> |
| <ul style="list-style-type: none"><li>• Co-directed a mentorship program guiding applicants through graduate admissions</li><li>• Coordinated mentor recruitment, matching, and program delivery across cohorts</li></ul>          |                            |
| <b>Secretary, Ghana Transportation Professionals Forum (GTPF)</b>                                                                                                                                                                  | <b>Feb 2026 – Present</b>  |
| <ul style="list-style-type: none"><li>• Maintain membership records and coordinate executive board communications</li><li>• Support event logistics and engagements with Ghana transportation stakeholders</li></ul>               |                            |

- Committee Member, Career Mixer Committee, CEE Dep't, Northeastern Univ.** **Sep 2025 – Feb 2026**
- Organizing networking events connecting students with industry
  - Coordinating logistics for career dev't workshops and employer engagements
- Executive C'ttee Member, CEE Graduate Student Council, Northeastern Univ.** **Feb 2024 – Present**
- Brainstorming and implementing strategies to enhance graduate student experience
  - Serving as a liaison between the departmental chair and the graduate student body
- Department Rep., College of Engineering. PhD Council, Northeastern Univ.** **Jun 2024 – Aug 2025**
- Meeting with advisor to discuss novel ways of improving student experience
  - Organizing weekly coffee hours and semesterly happy hours for PhD students
- Communications Director, Young Professionals and Youth Coalition, Ghana** **Jan 2021 – Dec 2021**
- Social media management to curate consistent communications content
  - Organization of in-person and virtual engagements with members and partners
- President, Civil Engineering Students' Association, KNUST** **Feb 2020 – Sep 2021**
- Coordination of fundraising events to secure administrative equipment
  - Establishment of academic support scheme for underachieving students

## SELECTED CONFERENCE PRESENTATIONS

- 
- |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | <ol style="list-style-type: none"> <li>1. <b>Obeng-Amoako, N. D.</b>, Furth, P., Akhavan, A., Sadeghinassr, B., Saeidi, R., Miranda-Moreno, L., Théberge-Barrette, C., &amp; Moser, P. (2026, August). <i>Intersections as Barriers to Low-Stress Cycling: Intersection Level of Traffic Stress and Dual-Graph Network Modeling</i>. Bridging Transportation Researchers (BTR) Conference, Online.</li> <li>2. Furth, P., &amp; <b>Obeng-Amoako N. D.</b> (2026, May). <i>Why Johnny and Mary Can't Ride to School: Identifying Barriers to Safe Bicycle Access</i>. Boston Area Research Initiative (BARI) Conference, Boston, MA.</li> <li>3. Yang, Y., Ahammed, D. T., <b>Obeng-Amoako, N. D.</b>, Christofa, E., Ai, C., &amp; Furth, P. (2026, January). <i>Community-Engaged Informed Floating Bus Stop Guidelines</i> [Poster]. Transportation Research Board Annual Meeting, Washington, DC.</li> </ol> |
| 2025 | <ol style="list-style-type: none"> <li>4. <b>Obeng-Amoako, N. D.</b>, &amp; Furth, P. (2025, May). <i>Underlying Traffic Signal Control: The Key to Effective Transit Signal Priority – A Case Study of MBTA Bus Route 39 along South Huntington Avenue, Boston, MA</i> [Poster]. MassDOT Transportation Innovation Conference, Worcester, MA.</li> <li>5. Yang, Y., Ahammed, D. T., <b>Obeng-Amoako, N. D.</b>, Christofa, E., Ai, C., &amp; Furth, P. (2025, June). <i>Design Guidelines for Accessible Floating Bus Stops</i>. International Conference on Transportation and Development (ASCE), Glendale, AZ.</li> </ol>                                                                                                                                                                                                                                                                                   |

- 2024      6. Yang, Y., Ahammed, D. T., **Obeng-Amoako, N. D.**, Christofa, E., Ai, C., & Furth, P. (2024, April). *Accessible Bus Stop Design in the Presence of Bike Lanes* [Poster]. MassDOT Transportation Innovation Conference, Worcester, MA.
- 2023      7. **Obeng-Amoako, N. D.**, Senaya, A. L., Kassim, B., Owusu-Ansah, F., Damoah, E., Worclachie, S., & Tutu, K. A. (2023, April). *Modification of Bitumen with Waste Materials for Enhanced Aggregate Retention in Surface-Dressed Roads*. 3rd IRF Africa Regional Congress & Exhibition, Accra, Ghana.

## TEACHING EXPERIENCE

---

### Northeastern University, Boston, MA

- Spring 2026      **Graduate Teaching Assistant, Energy Systems: Science, Technology, and Sustainability**
- Provided grading support for coursework on energy technologies and sustainability
  - Gave an invited lecture on Transportation Energy Systems in a Developing World
- Spring 2026      **Graduate Teaching Assistant, Transportation Systems Analysis and Planning**
- Supported students during office hours with questions on design software (Synchro)
  - Evaluated assignments on intersection analysis, signal timing, and street design principles
- Fall 2025      **Graduate Teaching Assistant, Highway Design**
- Provided grading support for homework assignments
  - Assisted students during office hours, addressing questions about design principles
- Fall 2025      **Graduate Teaching Assistant, Transportation Systems Analysis and Planning**
- Evaluated coursework and provided constructive feedback on analytical assignments
  - Supported students' understanding of transportation system planning frameworks
- Fall 2023      **Course Instructor, Recitation for Statics and Solid Mechanics**
- Tutored fundamental concepts of Newtonian physics and solid mechanics
  - Evaluated students' test performance and offered comprehensive feedback

## PROFESSIONAL AFFILIATIONS

---

- |                                                             |                           |
|-------------------------------------------------------------|---------------------------|
| • American Society of Civil Engineers (ASCE)                | <b>Feb 2025 – Present</b> |
| • Boston Society of Civil Engineers Section (BSCES) of ASCE | <b>Feb 2025 – Present</b> |
| • New England Intelligent Transportation Society (NEITS)    | <b>Sep 2023 – Present</b> |
| • Institute of Transportation Engineers (ITE)               | <b>Jan 2024 – Present</b> |

## VOLUNTEER EXPERIENCE & COMMUNITY INVOLVEMENT

---

- **Judge**, Massachusetts Science and Engineering Fair, Worcester (April 2025)
- **Cycling Activist**, Boston Cycling Union, Boston, MA (Feb 2025 – Present)
- **Executive Vice President**, Student Energy Society, KNUST (Jan 2022 – Oct 2022)
- **Community Outreach Coordinator**, Nat'l Society of Black Engineers, KNUST (Jan 2022 – Sep 2022)
- **Project Coordinator**, Engineers Without Borders, KNUST (Aug 2021 - Jul 2022)

## BLOG POSTS

---

1. “Ghana is Moving, but Barely: The Case for a Smarter Public Transportation Strategy.” MyJoyOnline, April 2026. <https://www.myjoyonline.com/ghana-is-moving-but-barely-the-case-for-a-smarter-public-transportation-strategy/>
2. “Replacing the Reid Overpass with an At-Grade Intersection (Cambridge, near BU Bridge).” Co-authored with Peter G. Furth. January 2026. <https://peterfurth.sites.northeastern.edu/2026/01/06/replacing-the-reid-overpass-with-an-at-grade-intersection-cambridge-near-bu-bridge/>
3. “PhD Student Reflects on the Sustainable Transportation Dialogue of Civilizations in Holland.” Northeastern University College of Engineering News, August 2024. <https://coe.northeastern.edu/news/a-reflection-on-the-sustainable-transportation-dialogue-of-civilizations-course-in-holland/>
4. “Bus-Friendly Traffic Signals Can Reduce Bus Delay By 90% on South Huntington Avenue.” Co-authored with Peter G. Furth. April 2024. <https://peterfurth.sites.northeastern.edu/2024/06/04/bus-friendly-traffic-signals-can-reduce-bus-delay-by-90-on-south-huntington-avenue/>

## REFERENCES

---

*Available upon request*